

Practical – 7

Aim: Given a grid of size N x M (N is the number of rows and M is the number of columns in the grid) consisting of '0's (Water) and '1's (Land). Find the number of islands.

Source Code :

```
#include <stdio.h>

#include <stdbool.h>

#include <string.h>

#define MAX 100

int rowNum[] = {-1, 0, 0, 1};

int colNum[] = {0, -1, 1, 0};

bool isSafe(int grid[MAX][MAX], int visited[MAX][MAX], int row, int col, int N, int M) {

    return (row >= 0) && (row < N) && (col >= 0) && (col < M) &&
    (grid[row][col] == 1 && !visited[row][col]);

}

void DFS(int grid[MAX][MAX], int visited[MAX][MAX], int row, int col, int N, int M) {

    visited[row][col] = 1;

    for (int k = 0; k < 4; ++k) {

        if (isSafe(grid, visited, row + rowNum[k], col + colNum[k], N, M)) {

            DFS(grid, visited, row + rowNum[k], col + colNum[k], N, M);

        }

    }

}
```

```
int countIslands(int grid[MAX][MAX], int N, int M) {
    int visited[MAX][MAX];
    memset(visited, 0, sizeof(visited));
    int count = 0;
    for (int i = 0; i < N; ++i) {
        for (int j = 0; j < M; ++j) {
            if (grid[i][j] == 1 && !visited[i][j]) {
                DFS(grid, visited, i, j, N, M);
                ++count;
            }
        }
    }
    return count;
}

int main() {
    int grid[MAX][MAX], N, M;
    printf("Enter the number of rows (N): \n");
    scanf("%d", &N);
    printf("Enter the number of columns (M): \n");
    scanf("%d", &M);
    printf("Enter the grid elements (0 for Water, 1 for Land):\n");
    for (int i = 0; i < N; ++i) {
        for (int j = 0; j < M; ++j) {
            scanf("%d", &grid[i][j]);
        }
    }
}
```

```

    }

    printf("Number of islands is: %d\n", countIslands(grid, N, M));

    return 0;
}

```

Output :-

Input :-

Enter the number of rows (N): 5

Enter the number of columns (M): 5

Enter the grid elements (0 for Water, 1 for Land):

1 0 1 0 1

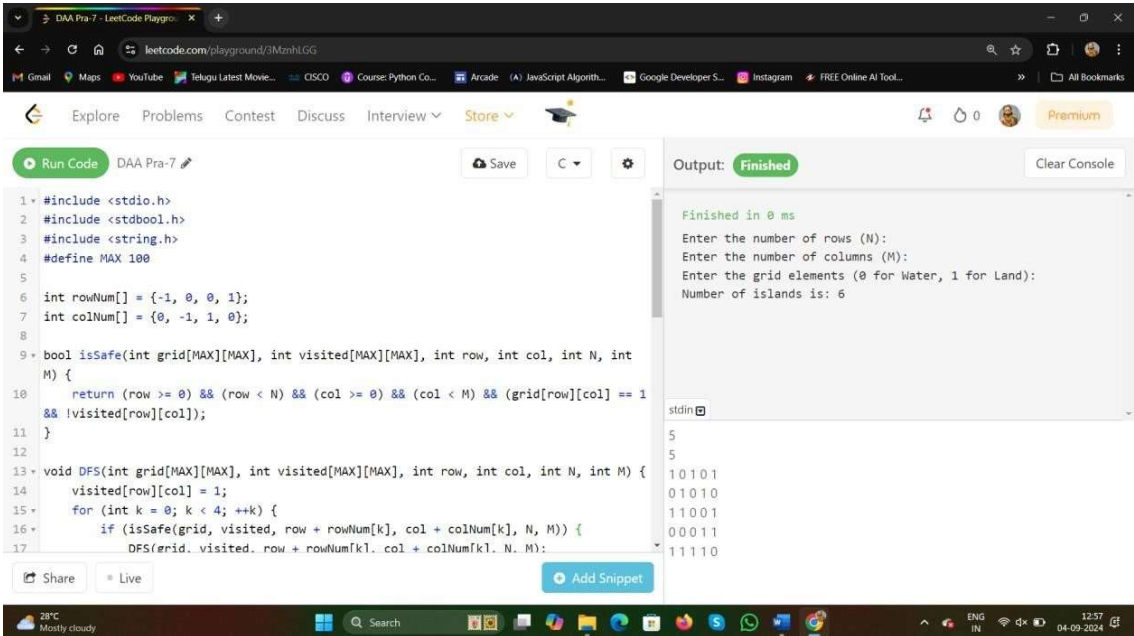
0 1 0 1 0

1 1 0 0 1

0 0 0 1 1

1 1 1 1 0

Output :- Number of islands is: 6



The screenshot shows a web-based C++ IDE with the following code and output:

```

1 #include <stdio.h>
2 #include <stdbool.h>
3 #include <string.h>
4 #define MAX 100
5
6 int rowNum[] = {-1, 0, 0, 1};
7 int colNum[] = {0, -1, 1, 0};
8
9 bool isSafe(int grid[MAX][MAX], int visited[MAX][MAX], int row, int col, int N, int M) {
10     return (row >= 0) && (row < N) && (col >= 0) && (col < M) && (grid[row][col] == 1
11     && !visited[row][col]);
12 }
13
14 void DFS(int grid[MAX][MAX], int visited[MAX][MAX], int row, int col, int N, int M) {
15     visited[row][col] = 1;
16     for (int k = 0; k < 4; ++k) {
17         if (isSafe(grid, visited, row + rowNum[k], col + colNum[k], N, M)) {
18             DFS(grid, visited, row + rowNum[k], col + colNum[k], N, M);
19         }
20     }
21 }
22
23 int main() {
24     int N, M;
25     scanf("%d %d", &N, &M);
26     int grid[N][M];
27     for (int i = 0; i < N; ++i) {
28         for (int j = 0; j < M; ++j) {
29             scanf("%d", &grid[i][j]);
30         }
31     }
32     int count = 0;
33     for (int i = 0; i < N; ++i) {
34         for (int j = 0; j < M; ++j) {
35             if (grid[i][j] == 1 && !visited[i][j]) {
36                 DFS(grid, visited, i, j, N, M);
37                 count++;
38             }
39         }
40     }
41     printf("Number of islands is: %d\n", count);
42     return 0;
43 }

```

Output: Finished in 0 ms

Enter the number of rows (N): 5

Enter the number of columns (M): 5

Enter the grid elements (0 for Water, 1 for Land):

Number of islands is: 6